# Higher dimensions

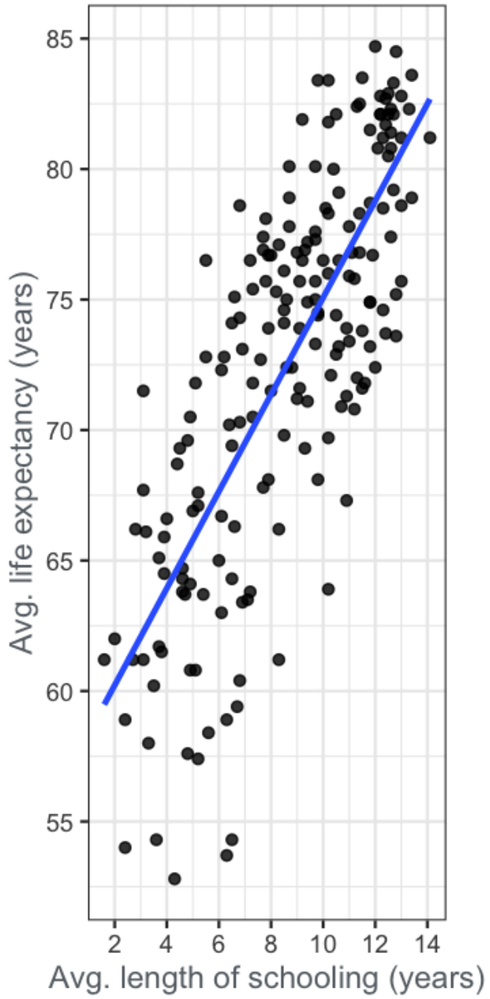
UNDERSTANDING DATA VISUALIZATION

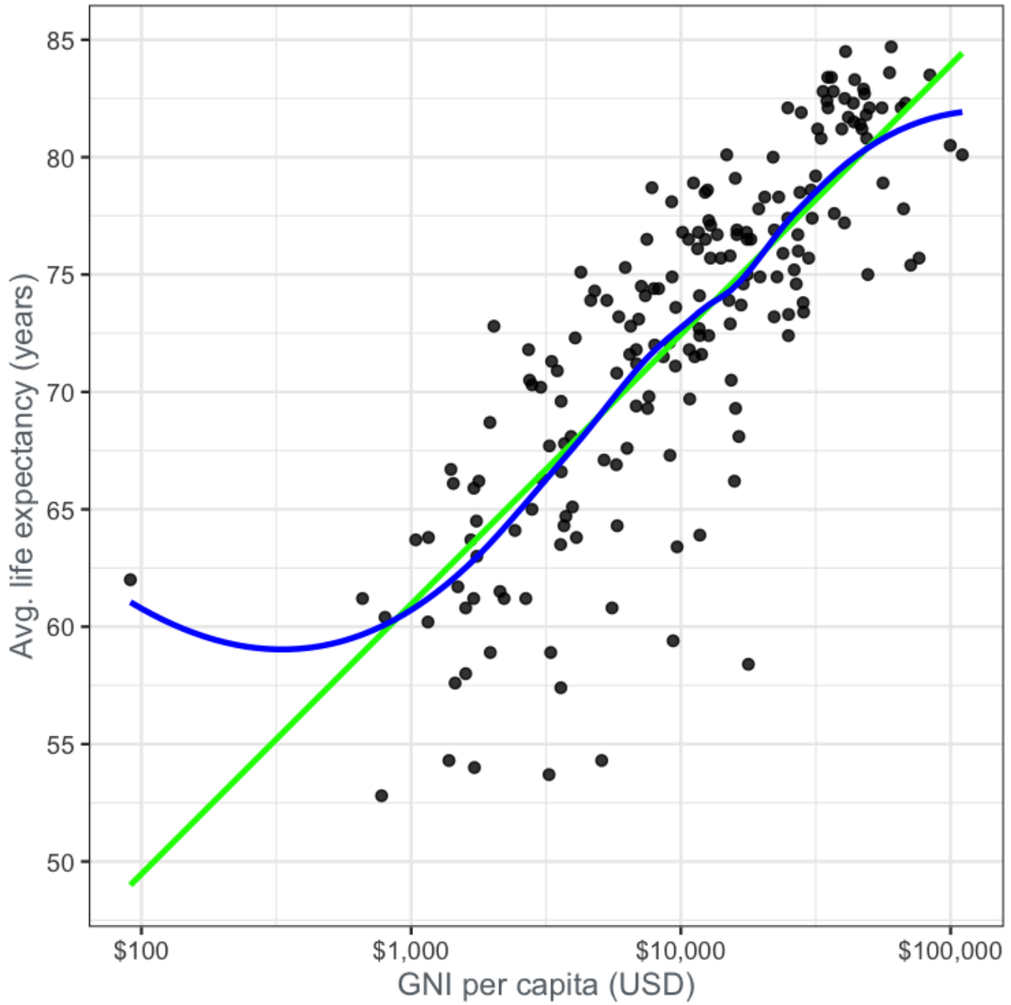


**Richie Cotton**
Data Evangelist at DataCamp

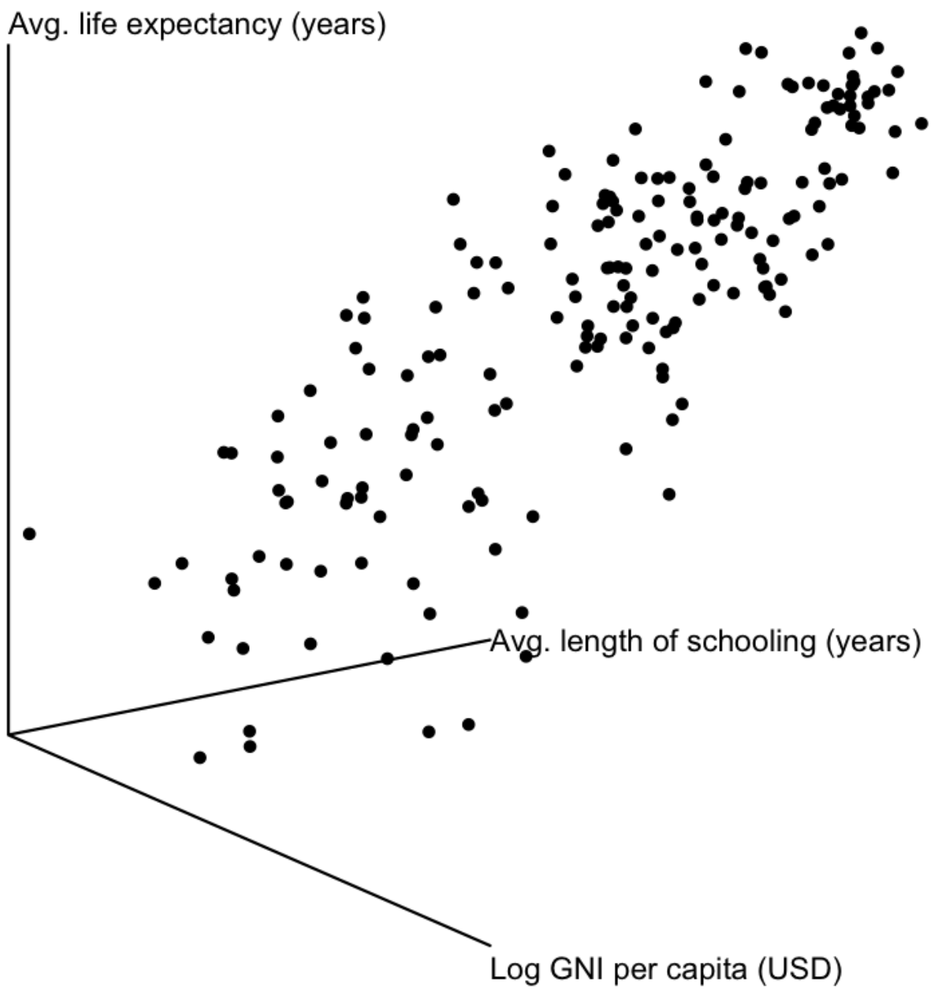
datacamp

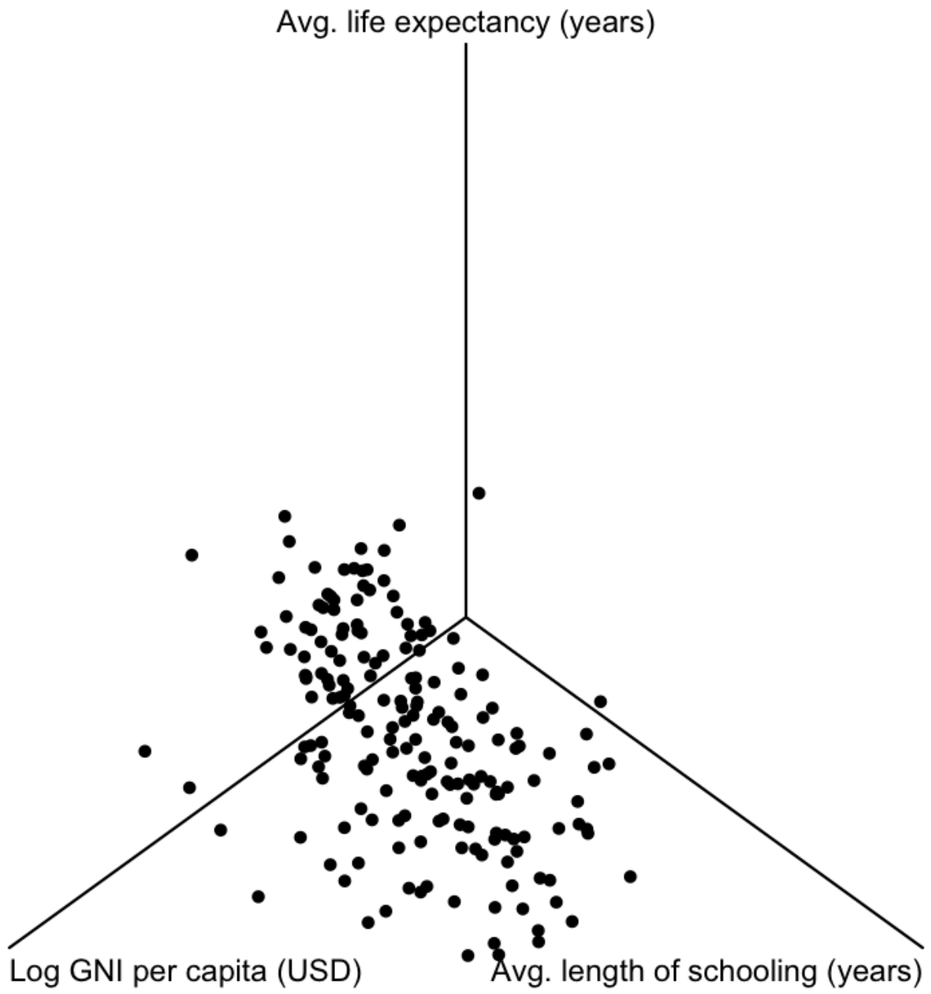
# The UN life expectancy scatter plots



# 3D scatter plots



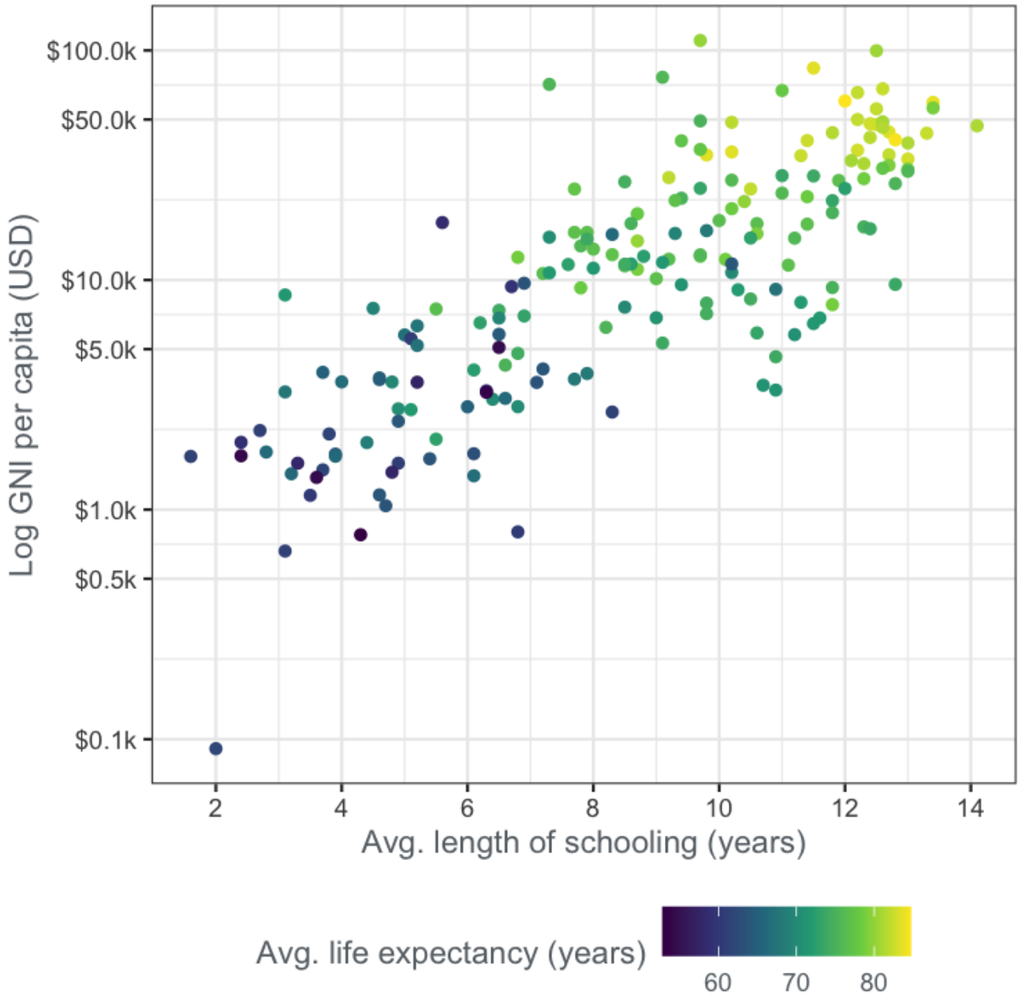



# x and y are not the only dimensions

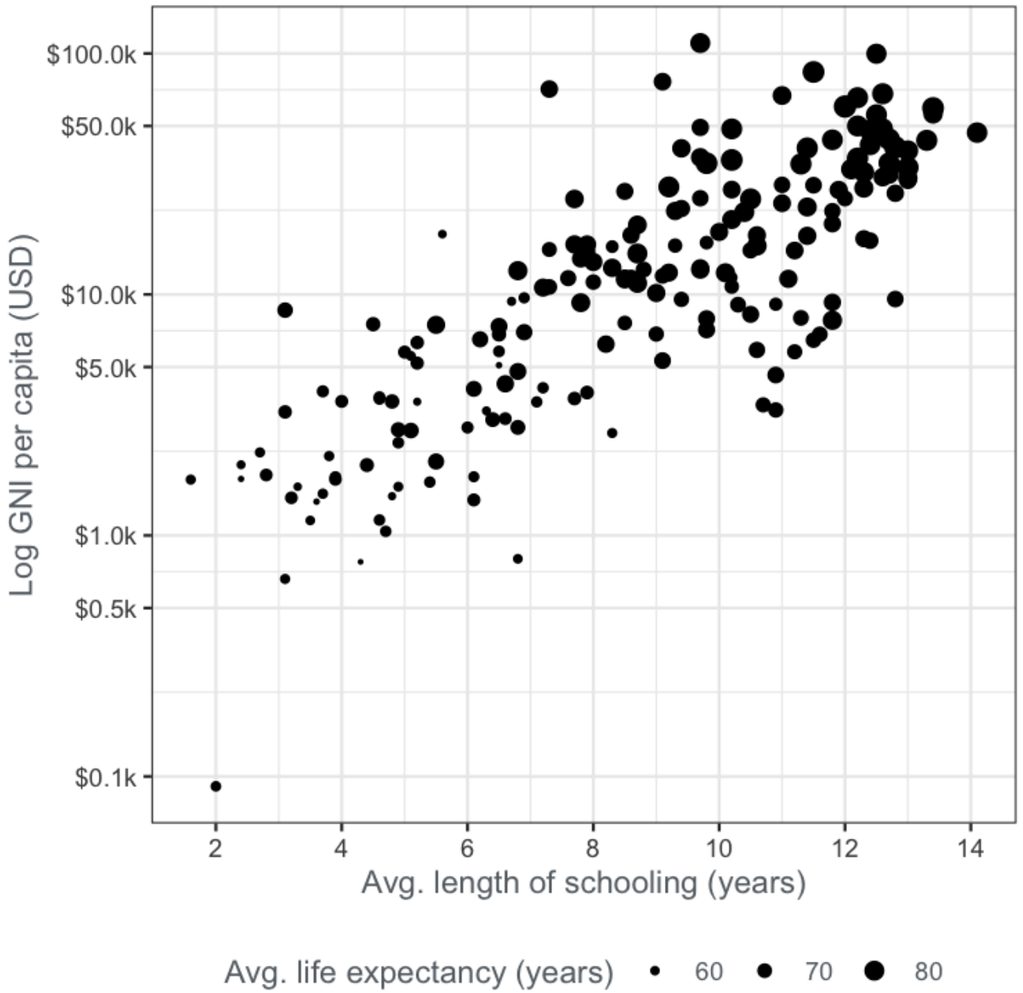
Points also have these dimensions

- color
- size
- transparency
- shape

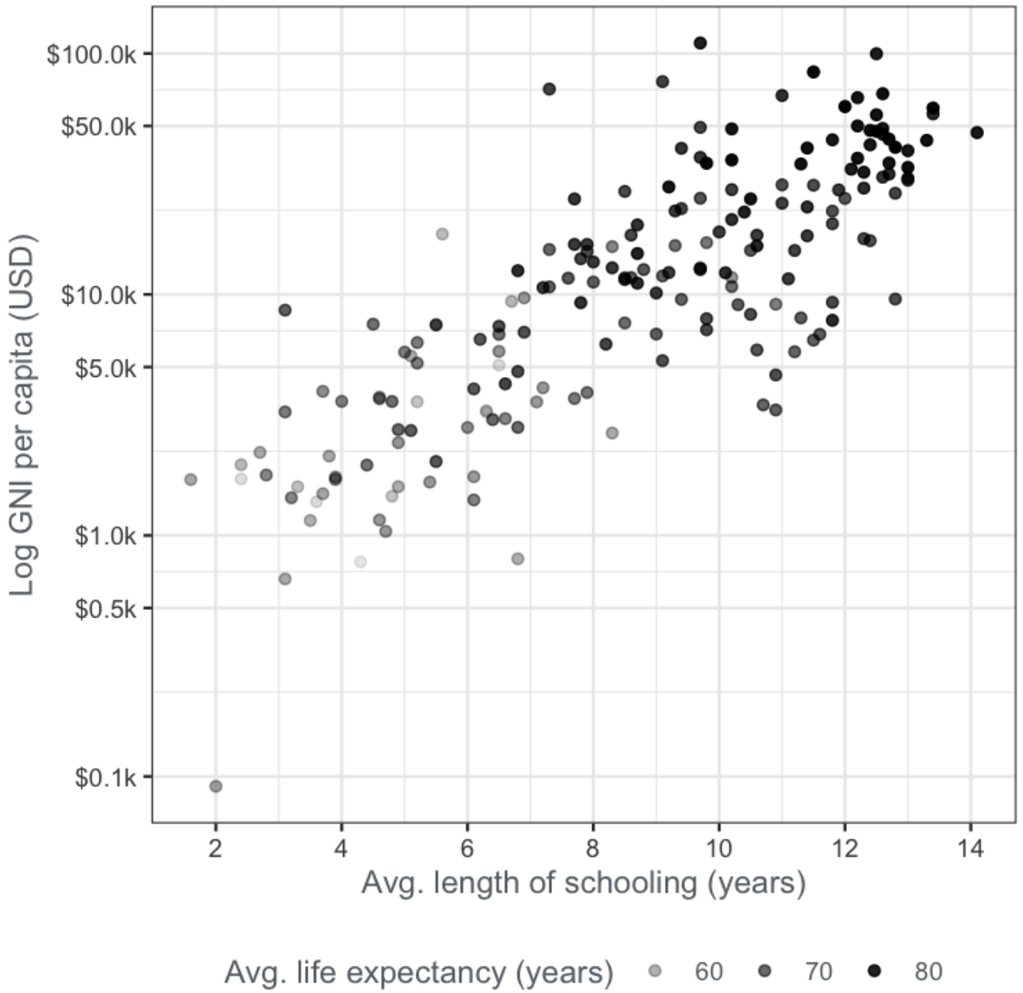
# Color


$100.0k
$50.0k
$10.0k
$5.0k
$1.0k
$0.5k
$0.1k
Log GNI per capita (USD)
2
4
6
8
10
12
14
Avg. length of schooling (years)
Avg. life expectancy (years)
60
70
80

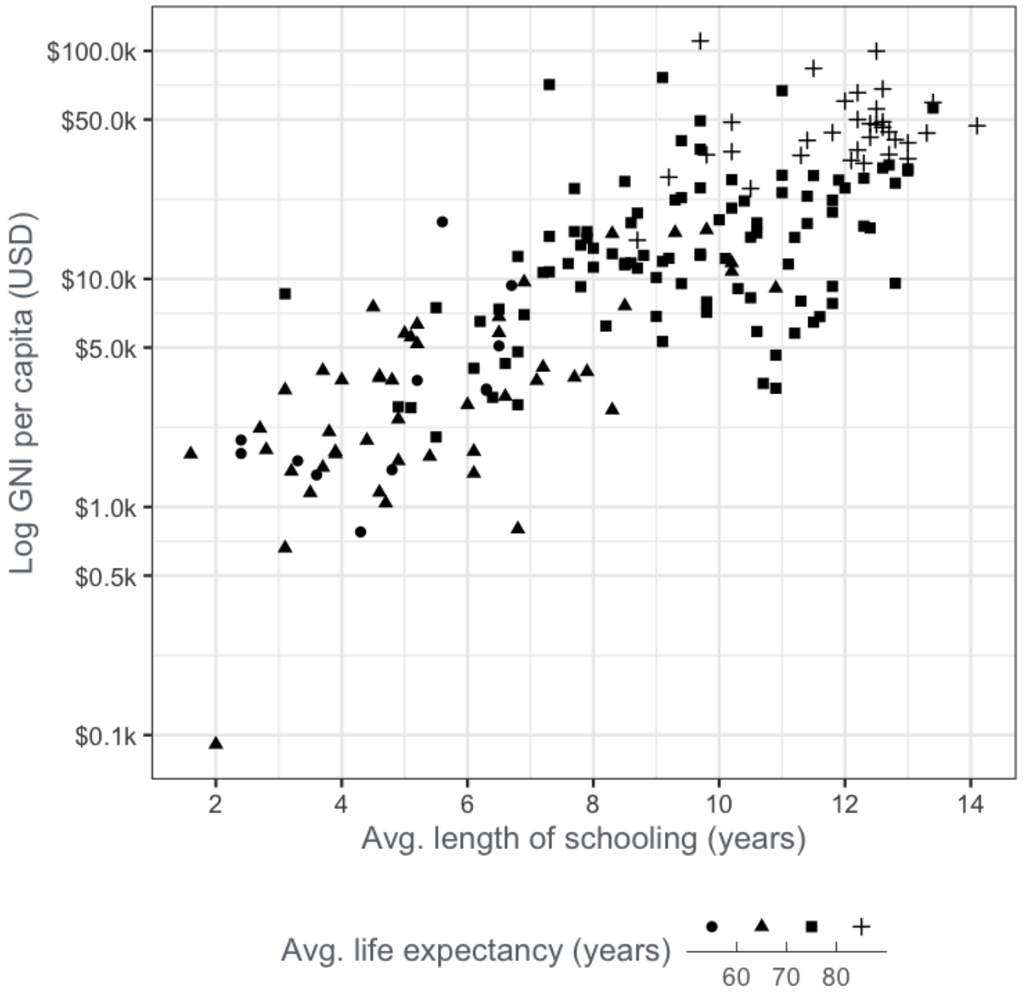
# Size


$100.0k
$50.0k
$10.0k
$5.0k
$1.0k
$0.5k
$0.1k
Log GNI per capita (USD)
2
4
6
8
10
12
14
Avg. length of schooling (years)
Avg. life expectancy (years)
60
70
80

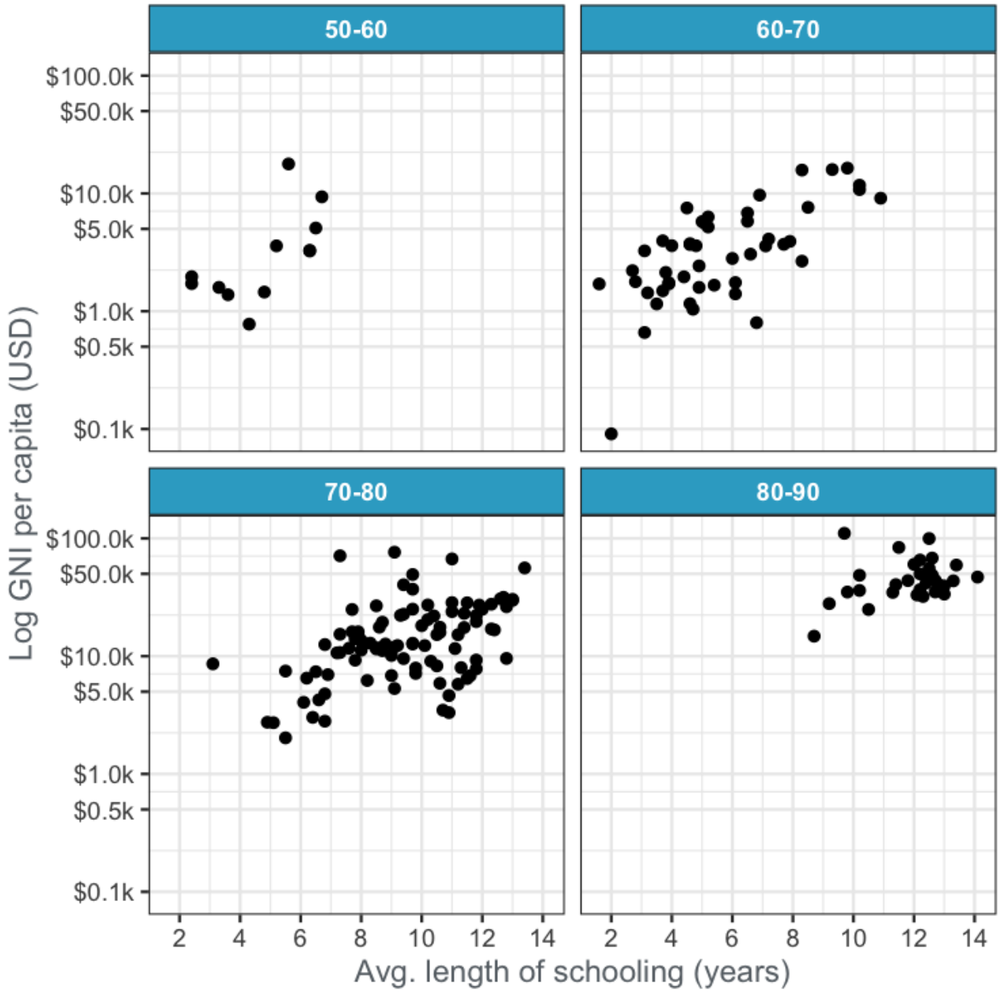
# Transparency



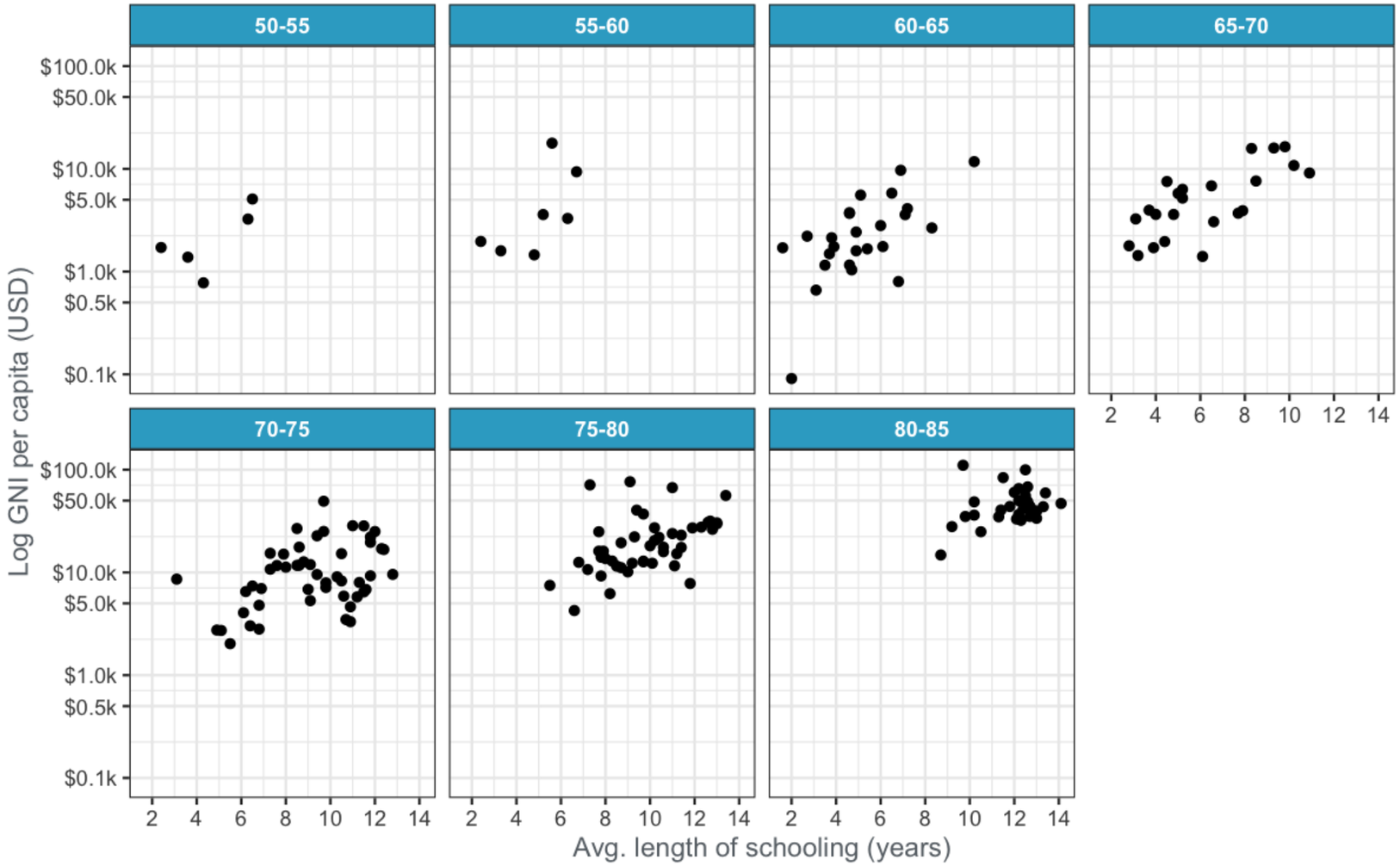
# Shape


$100.0k
$50.0k
$10.0k
$5.0k
$1.0k
$0.5k
$0.1k
Log GNI per capita (USD)
2
4
6
8
10
12
14
Avg. length of schooling (years)
Avg. life expectancy (years)
60
70
80

# Lots of panels



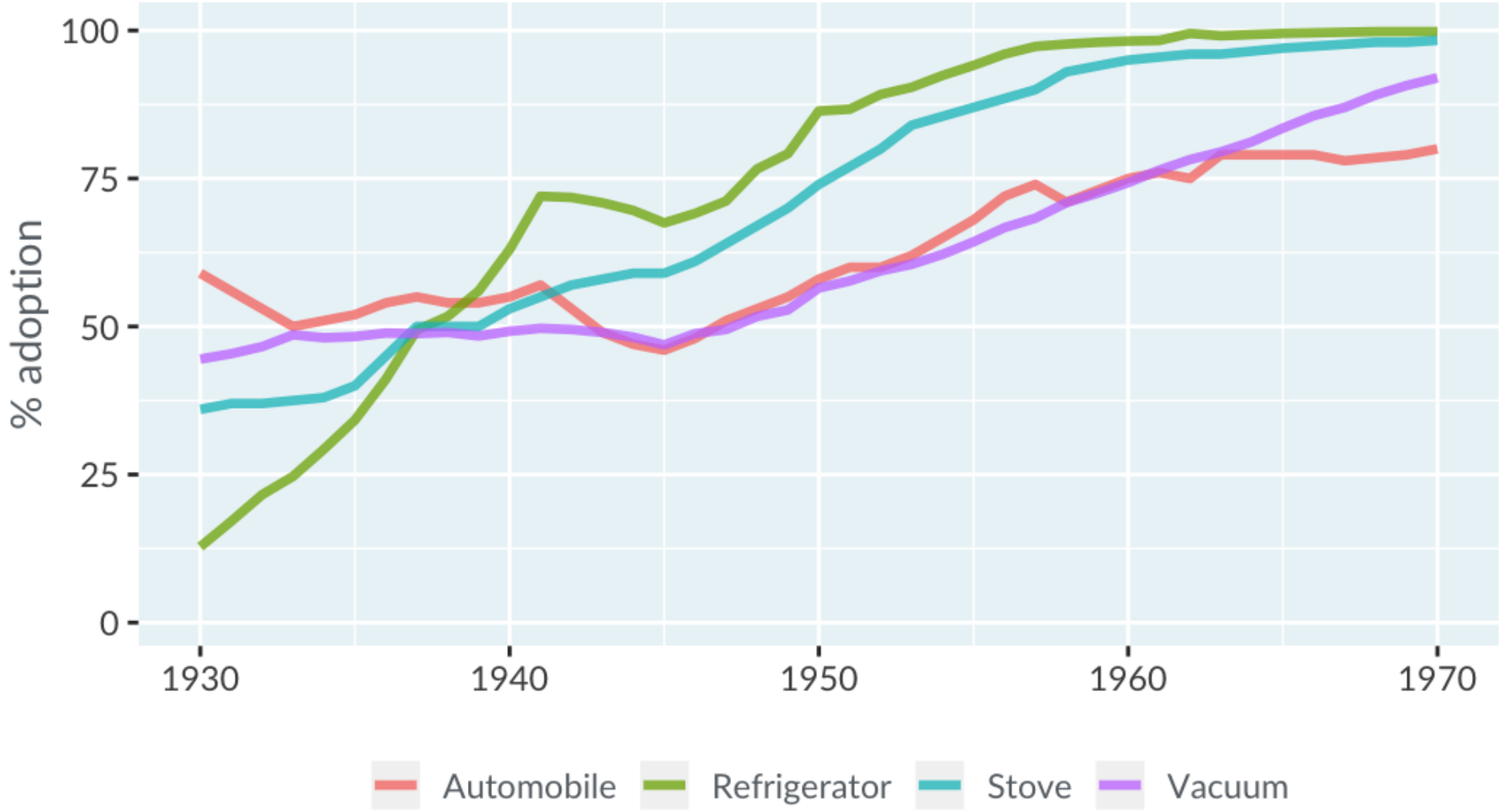
# Even more panels



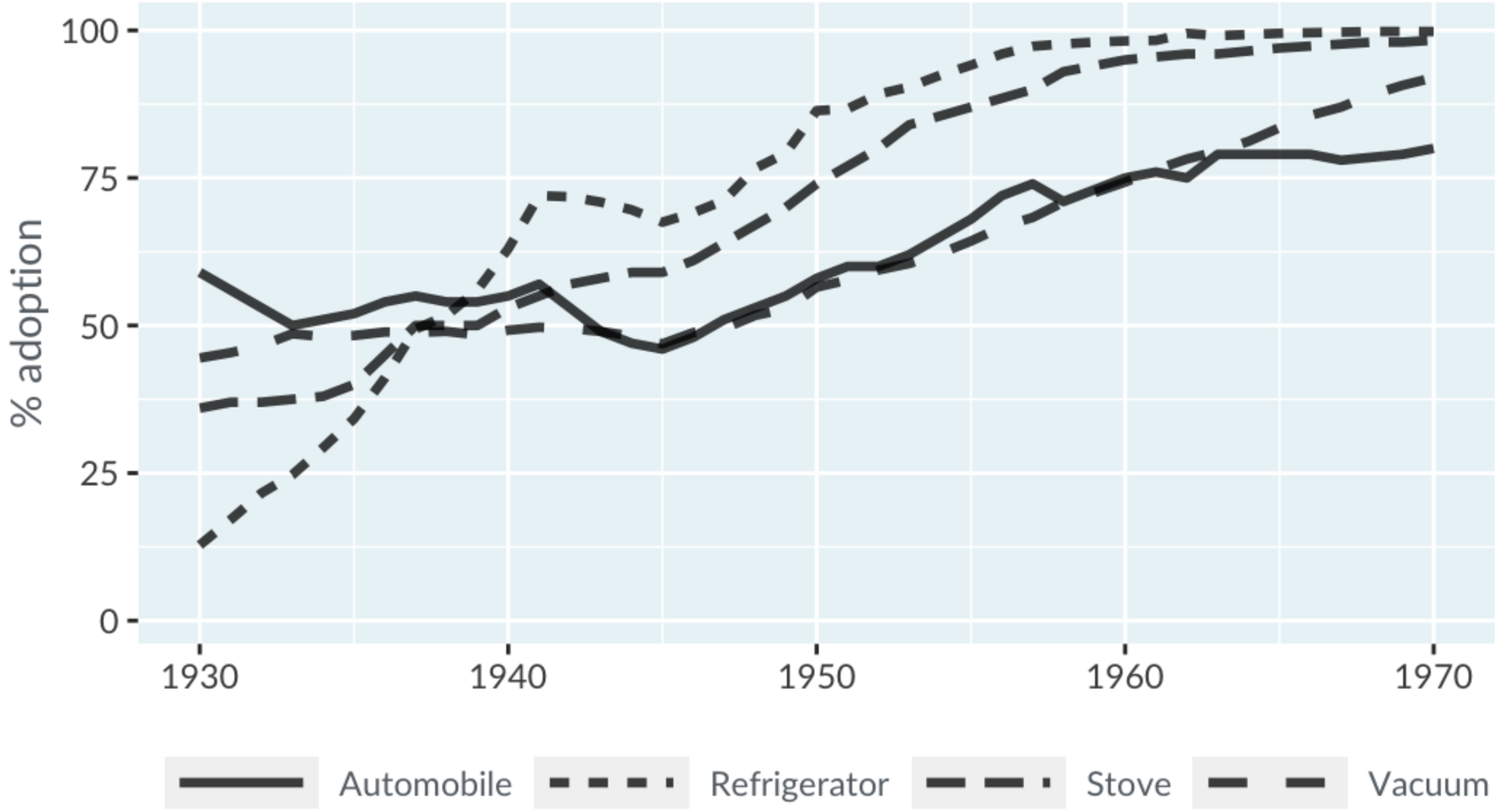
# Other dimensions for line plots

- color
- thickness
- transparency
- line type (solid, dashes, dots)

# Color


100
75
50
25
0
% adoption
1930
1940
1950
1960
1970
Automobile
Refrigerator
Stove
Vacuum

# Linetype


100
75
50
25
0
% adoption
1930
1940
1950
1960
1970
Automobile
Refrigerator
Stove
Vacuum

# Let's practice!

UNDERSTANDING DATA VISUALIZATION

# Using color

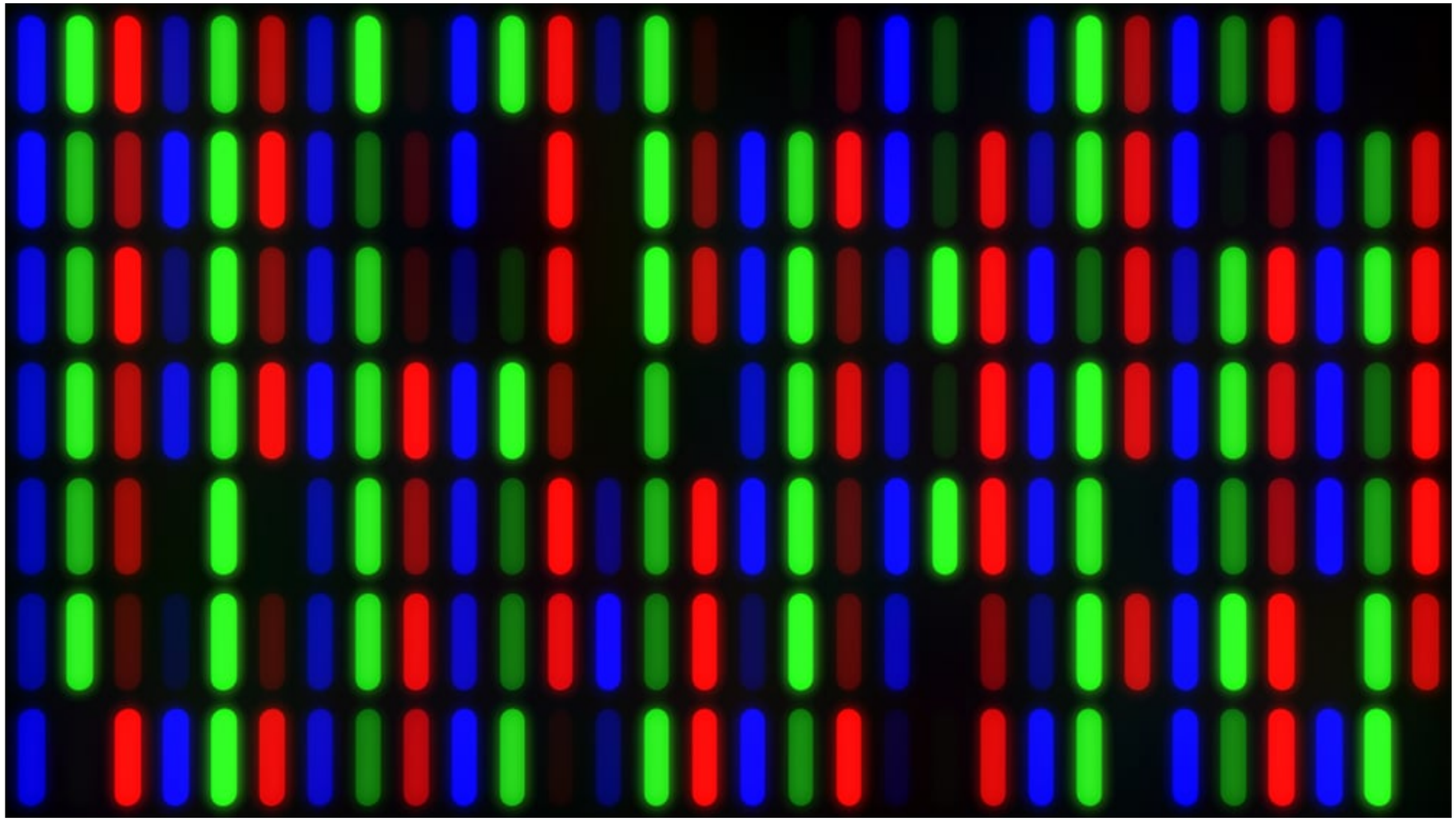
## UNDERSTANDING DATA VISUALIZATION



**Richie Cotton**
Data Evangelist at DataCamp

datacamp

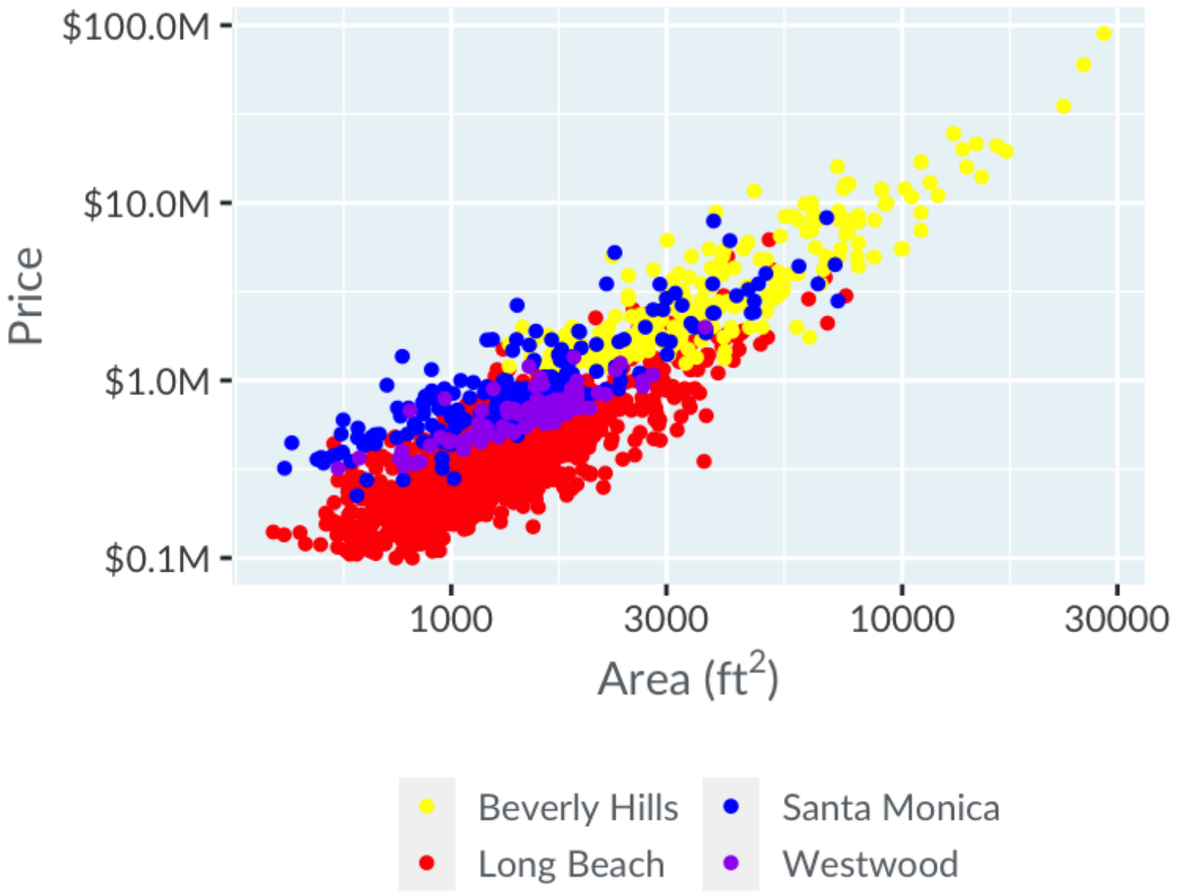
# Colorspaces: Red-Green-Blue



# Colorspaces: Cyan-Magenta-Yellow-blacK



# Choosing a plotting palette




- Usually, each color should stand out as much as other colors.
- The perceptual distance from one color in the plot to the next should be constant.

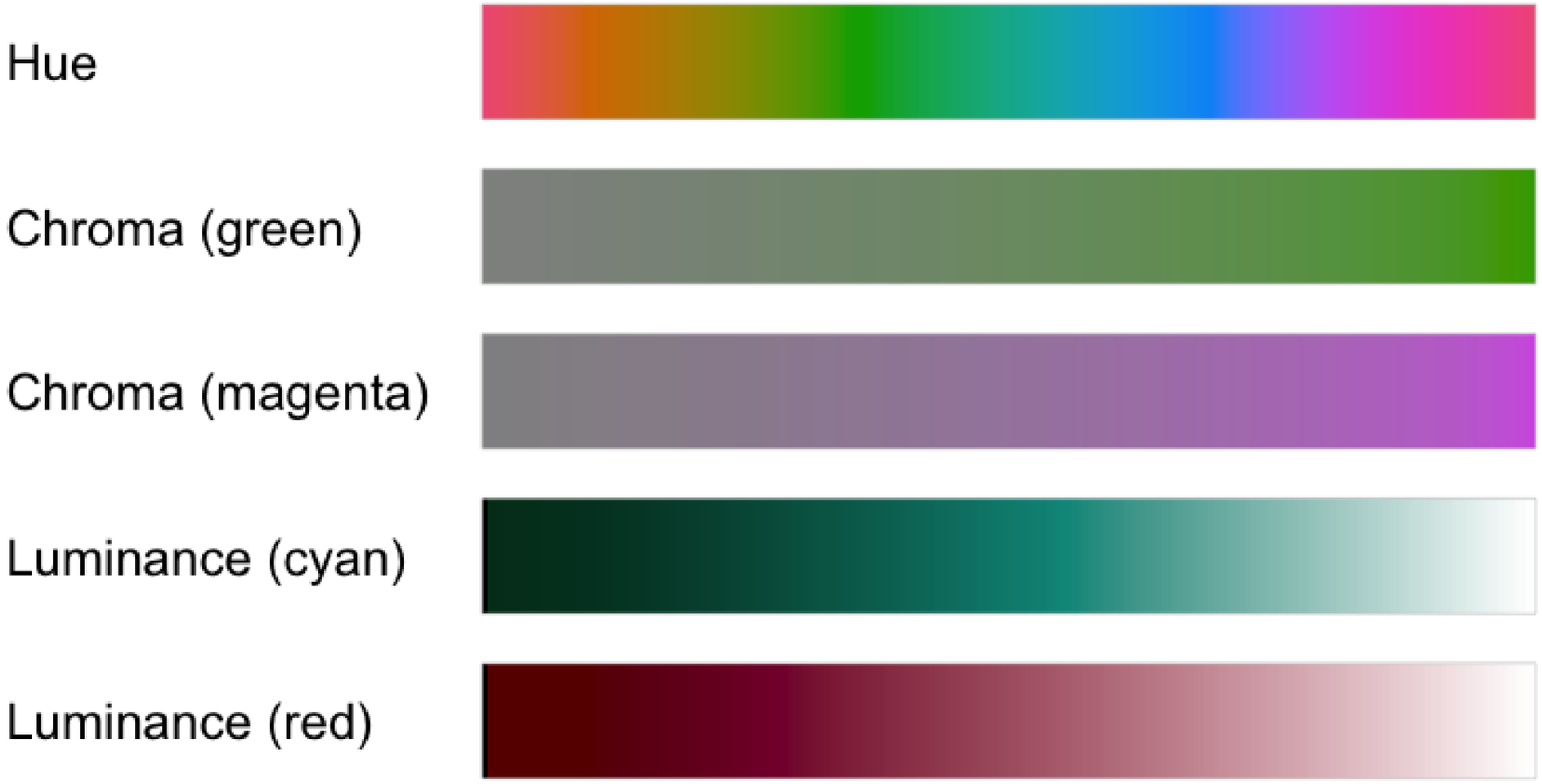
# Colorspaces: Hue-Chroma-Luminance



# Colorspaces: Hue-Chroma-Luminance



# Colorspaces: Hue-Chroma-Luminance

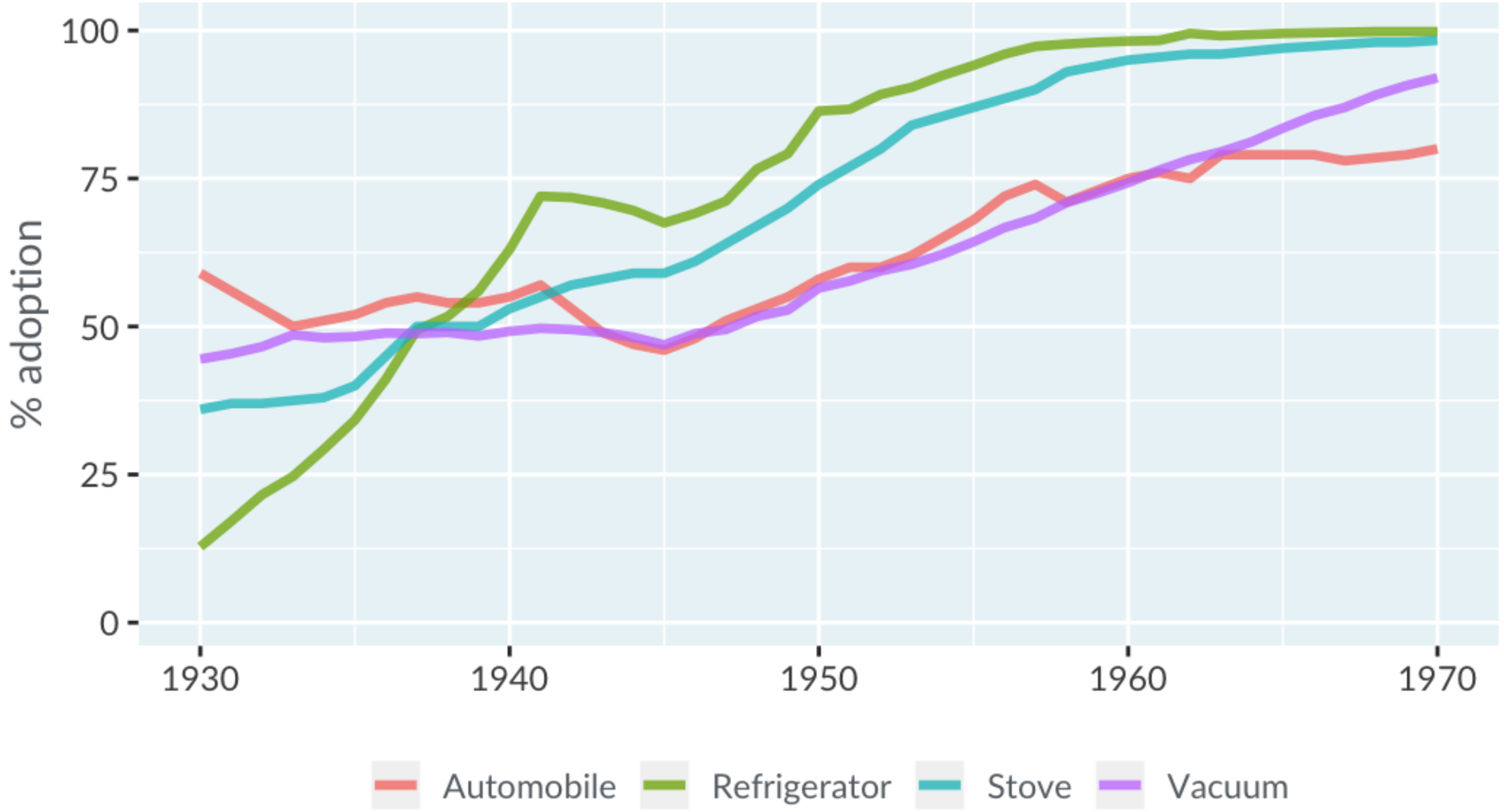


# Three types of color scale: qualitative

| Type | Purpose | What to vary |
| --- | --- | --- |
| qualitative | Distinguish unordered categories | hue |



# Qualitative palette example

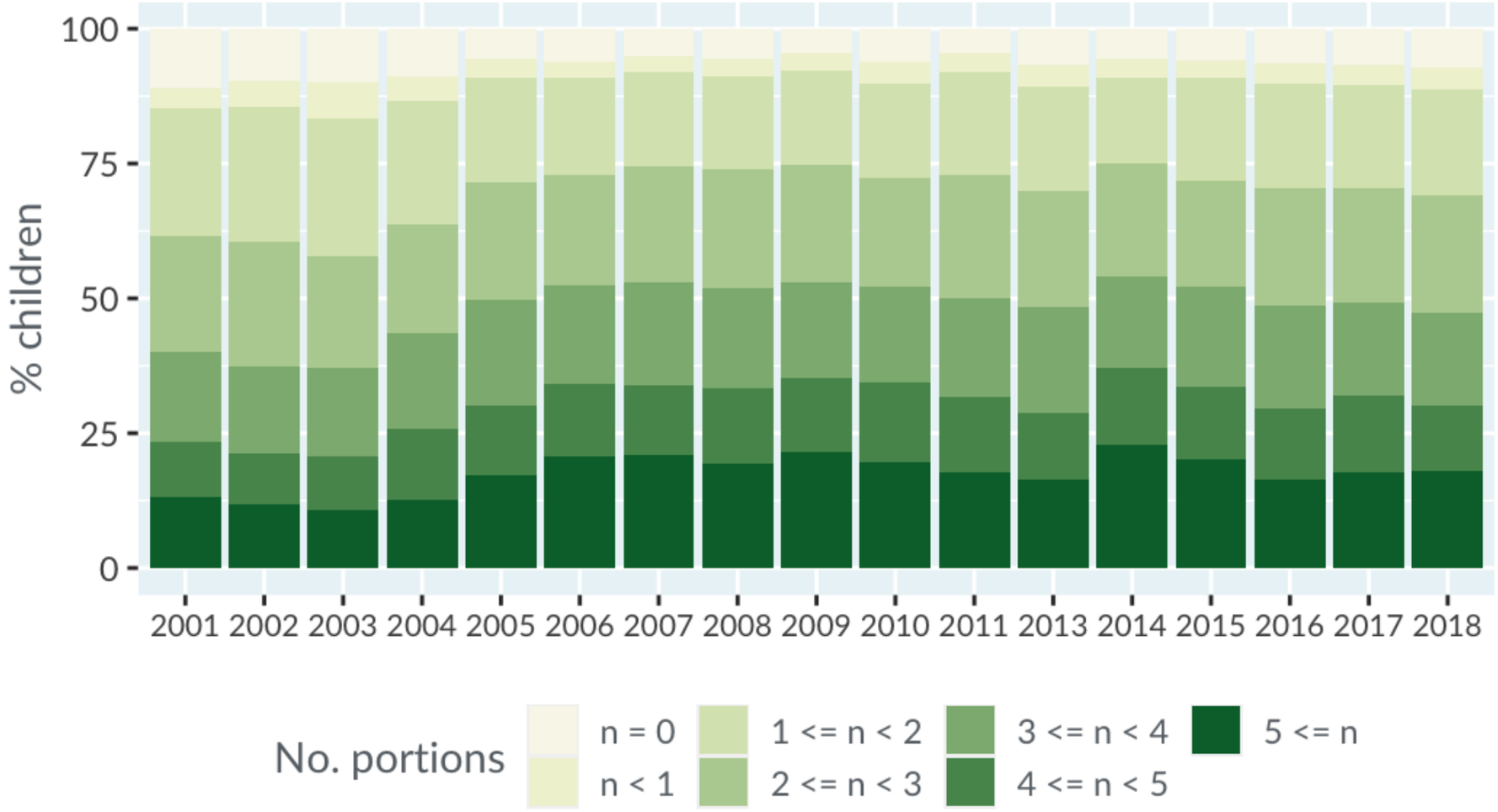


# Three types of color scale: sequential

| Type | Purpose | What to vary |
| --- | --- | --- |
| sequential | Show ordering | chroma or luminance |



# Sequential palette example



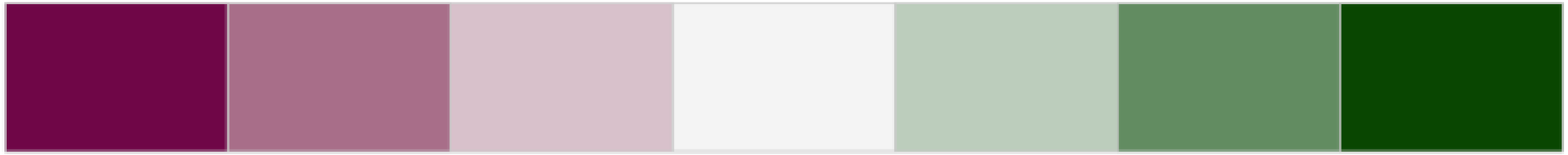
# Another sequential palette example




[1] Viridis color scale: https://bids.github.io/colormap

# Three types of color scale: diverging

| Type | Purpose | What to vary |
| --- | --- | --- |
| diverging | Show above or below a midpoint | chroma or luminance, with 2 hues |

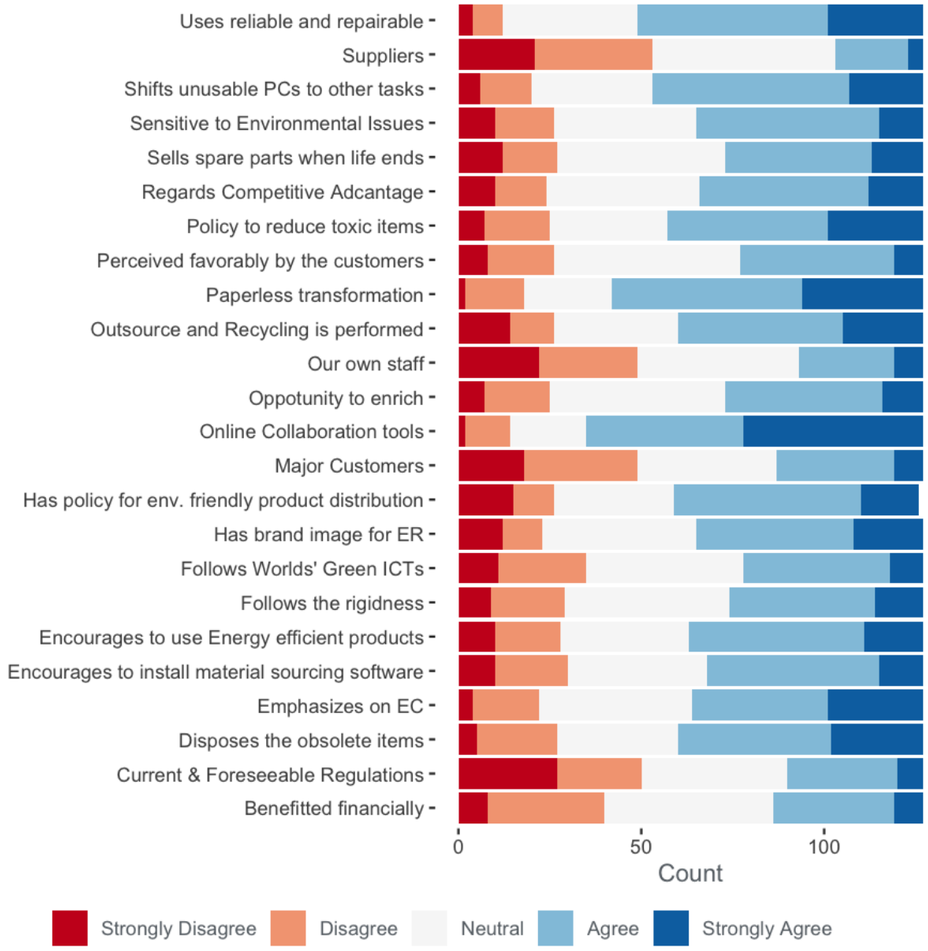


# Green Tech in Malaysia survey dataset

| question | response | n |
|---|---|---|
| Uses reliable and repairable | Strongly Disagree | 4 |
| Uses reliable and repairable | Disagree | 8 |
| Uses reliable and repairable | Neutral | 37 |
| Uses reliable and repairable | Agree | 52 |
| Uses reliable and repairable | Strongly Agree | 26 |
| ... | ... | ... |

[1] Islam et al. (2019) http://dx.doi.org/10.17632/wggvryfhsk.1

# Diverging palette example



# Let's practice!

UNDERSTANDING DATA VISUALIZATION

# Plotting many variables at once

UNDERSTANDING DATA VISUALIZATION

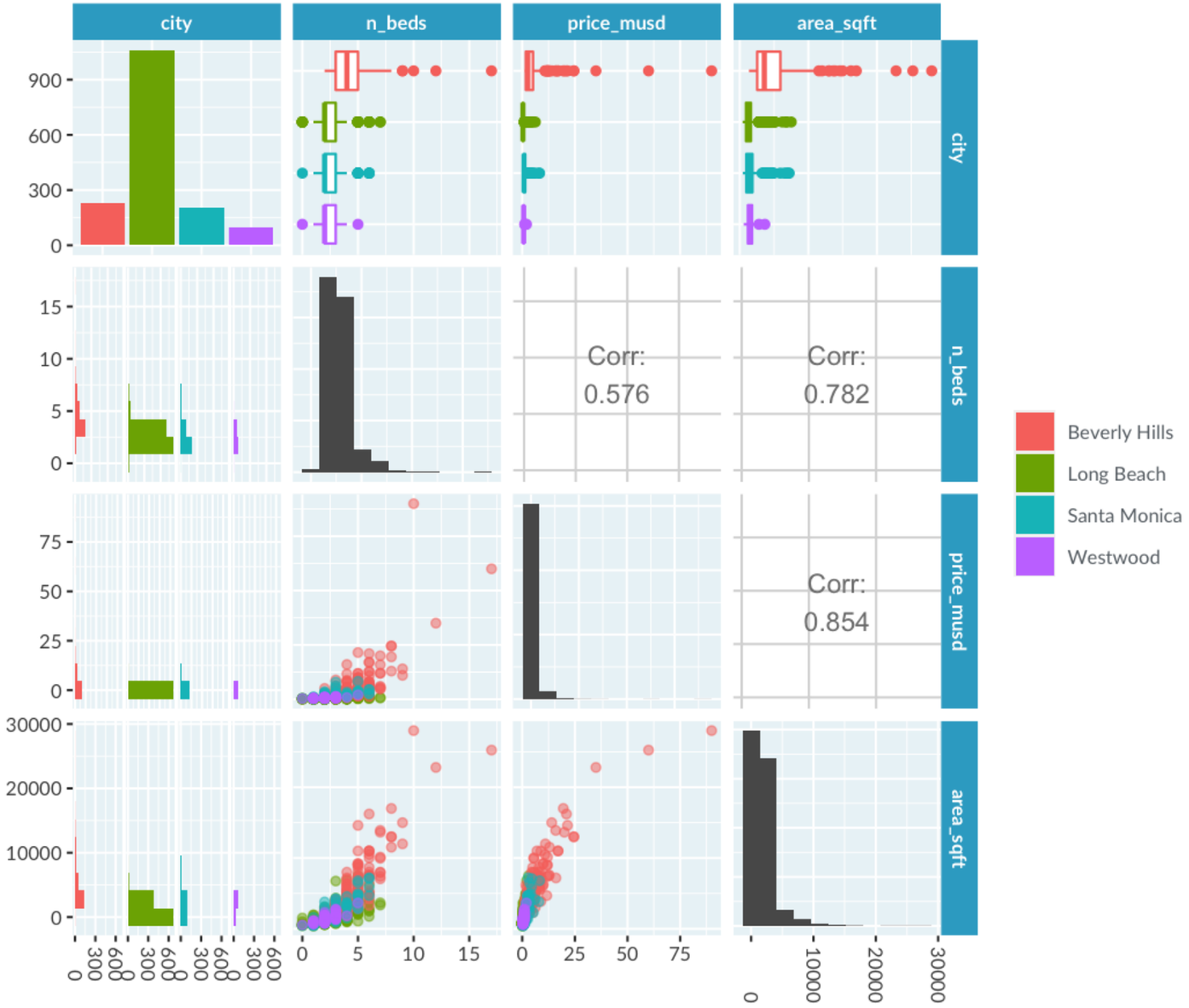


**Richie Cotton**
Data Evangelist

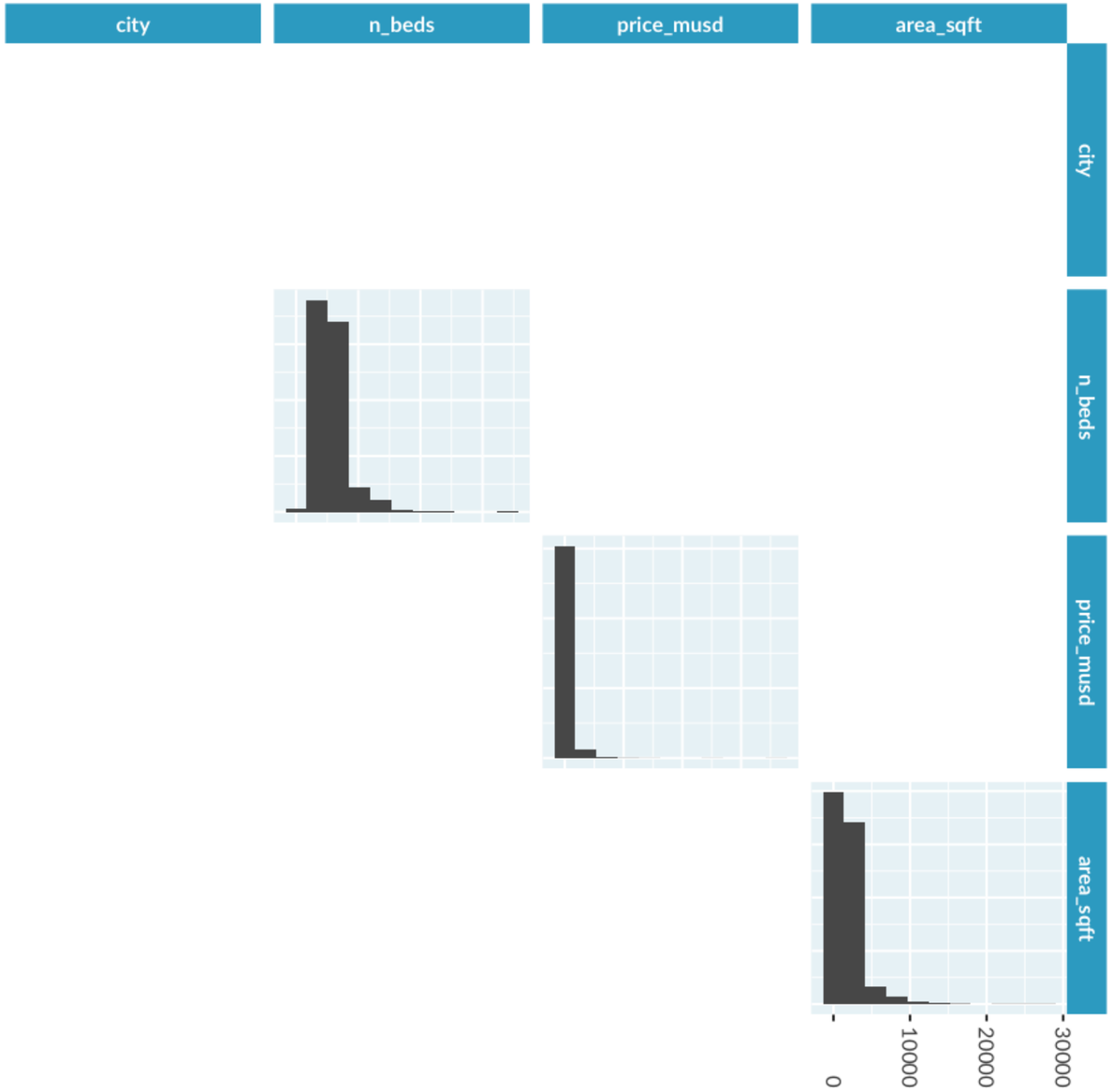
datacamp

# When should you use a pair plot?

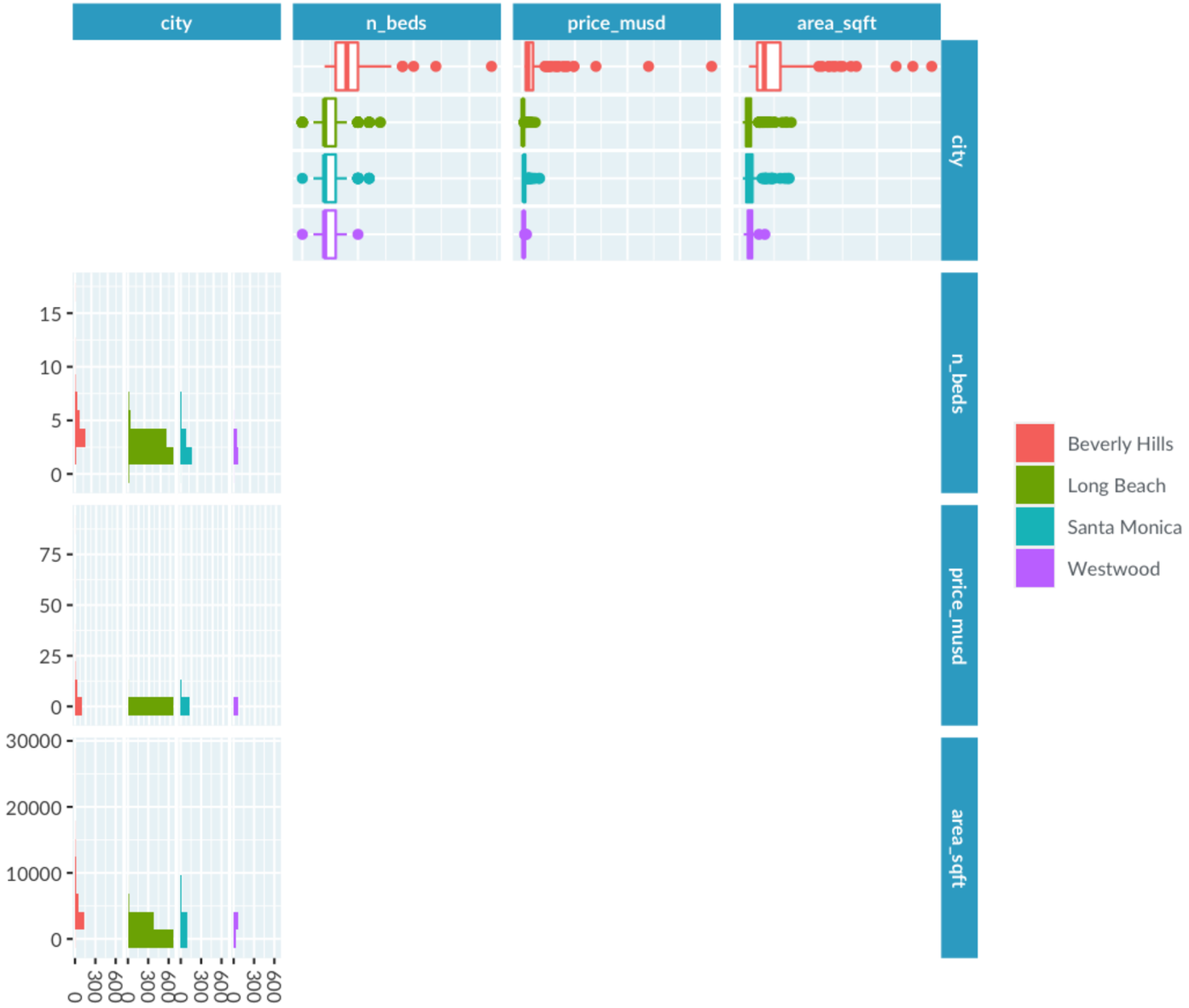
- You have up to ten variables (either continuous, categorical, or a mix).
- You want to see the distribution for each variable.
- You want to see the relationship between each pair of variables.

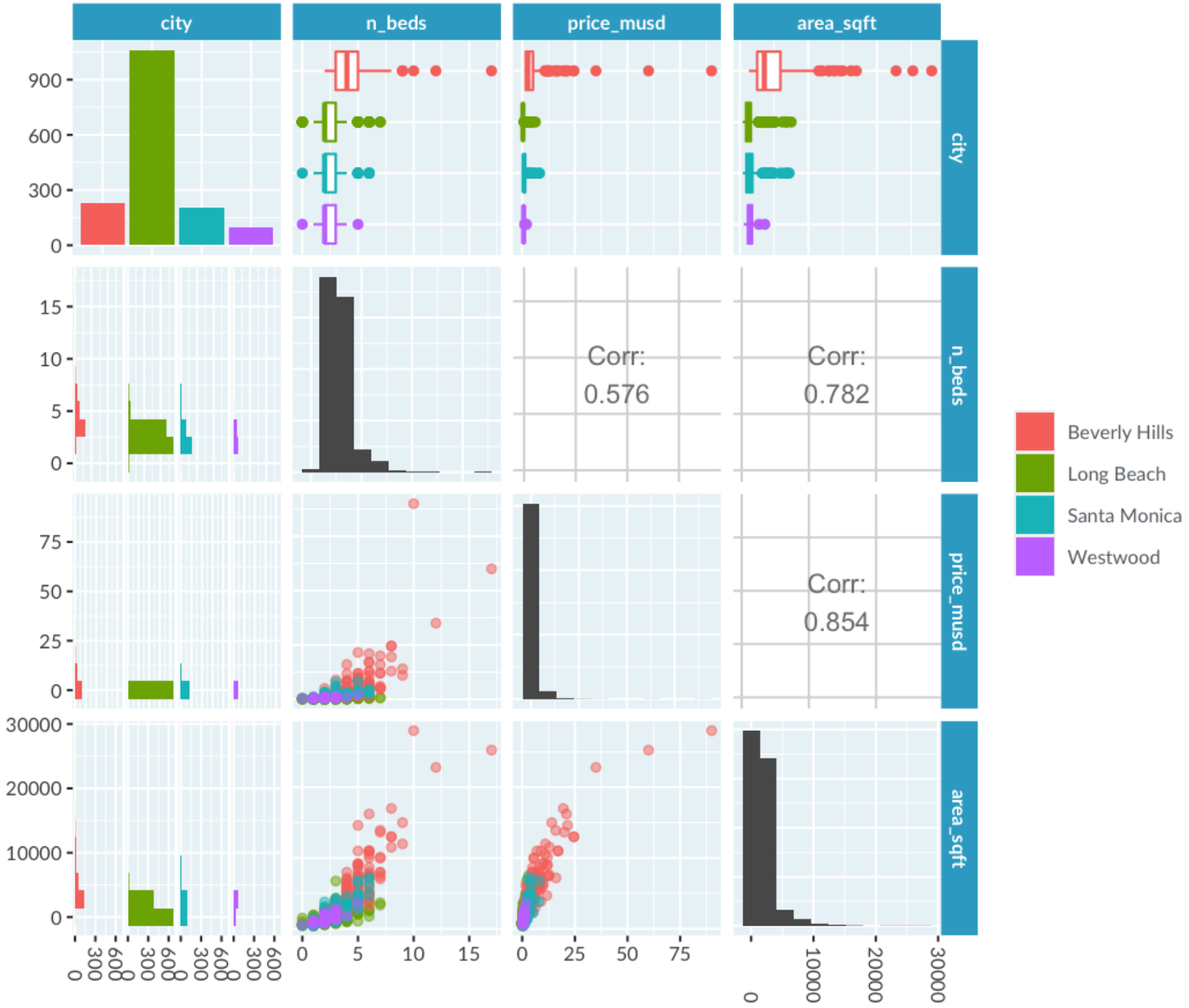

city
n_beds
price_musd
area_sqft
Corr:
0.576
Corr:
0.782
Corr:
0.854
Beverly Hills
Long Beach
Santa Monica
Westwood

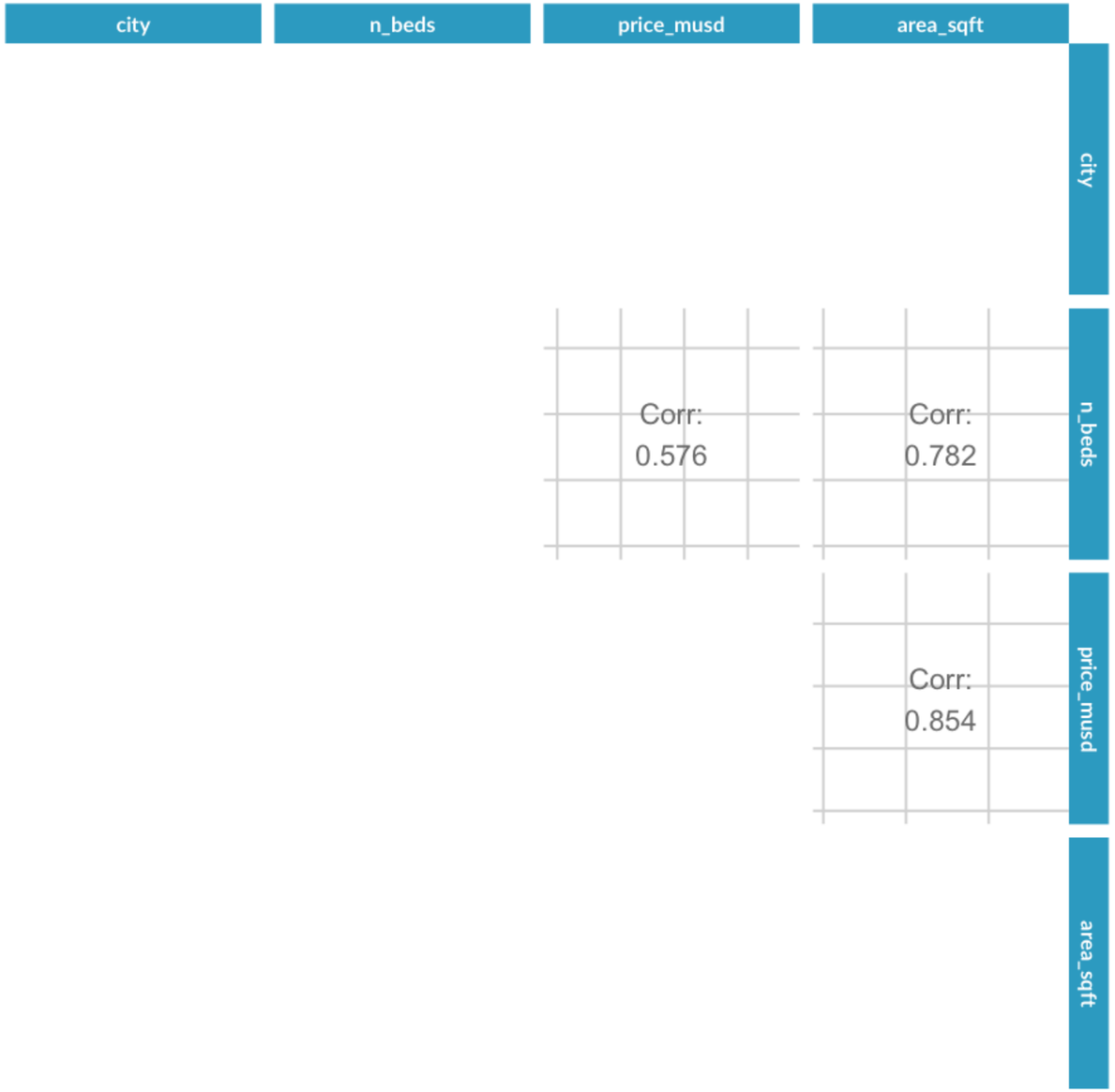

city
n_beds
price_musd
area_sqft
900
600
300
0
city
n_beds
price_musd
area_sqft
Beverly Hills
Long Beach
Santa Monica
Westwood


city
n_beds
price_musd
area_sqft
city
n_beds
price_musd
area_sqft
0
10000
20000
30000


city
n_beds
price_musd
area_sqft
Corr:
0.576
Corr:
0.782
Corr:
0.854
Beverly Hills
Long Beach
Santa Monica
Westwood
0
5
10
15
0
25
50
75

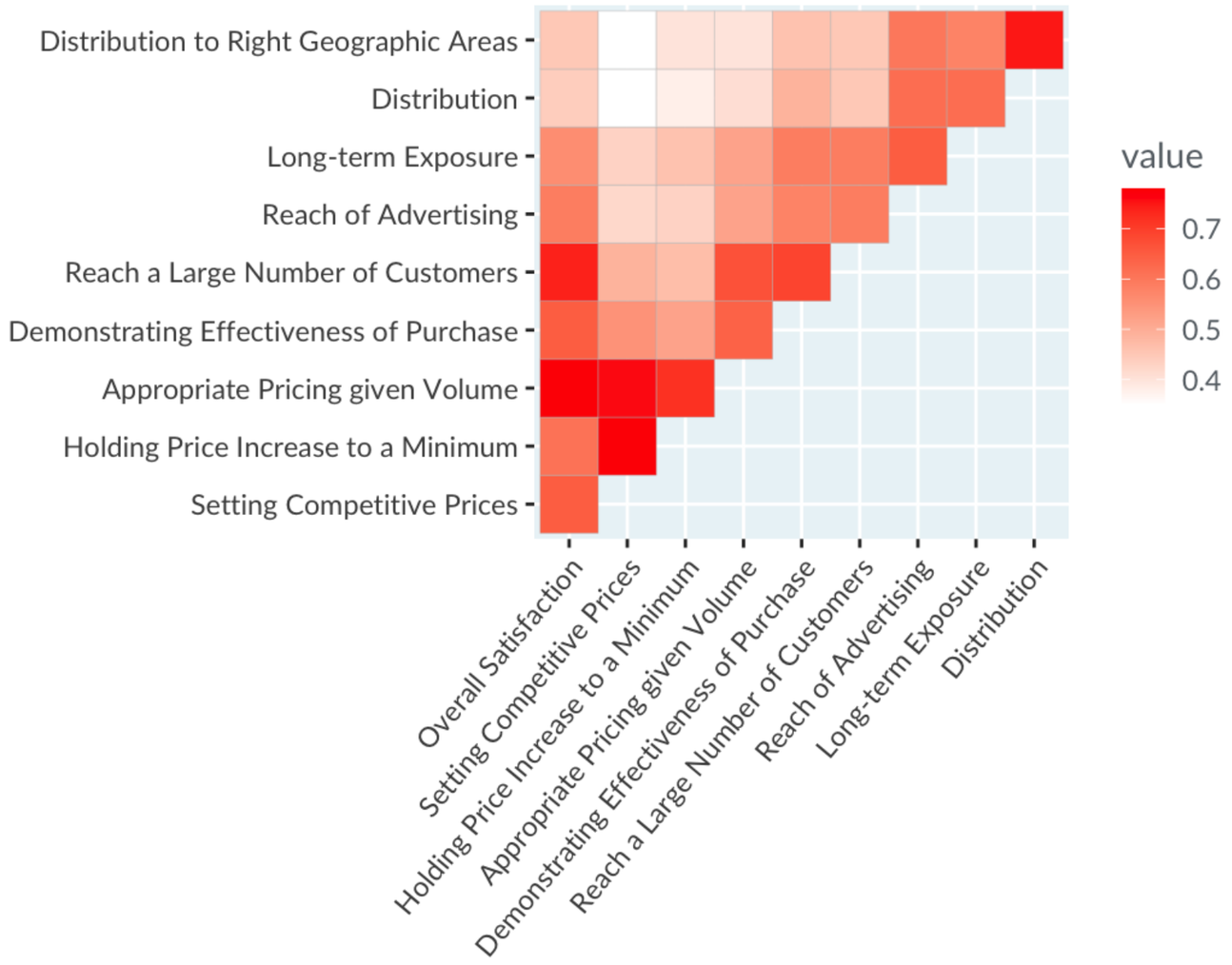

city
n_beds
price_musd
area_sqft
city
n_beds
price_musd
area_sqft
15
10
5
0
75
50
25
0
30000
20000
10000
0
0
300
600
900
300
600
900
300
600
900
300
600
Beverly Hills
Long Beach
Santa Monica
Westwood


city
n_beds
price_musd
area_sqft
Corr:
0.576
Corr:
0.782
Corr:
0.854
Beverly Hills
Long Beach
Santa Monica
Westwood


city
n_beds
price_musd
area_sqft
city
Corr:
0.576
Corr:
0.782
n_beds
Corr:
0.854
price_musd
area_sqft

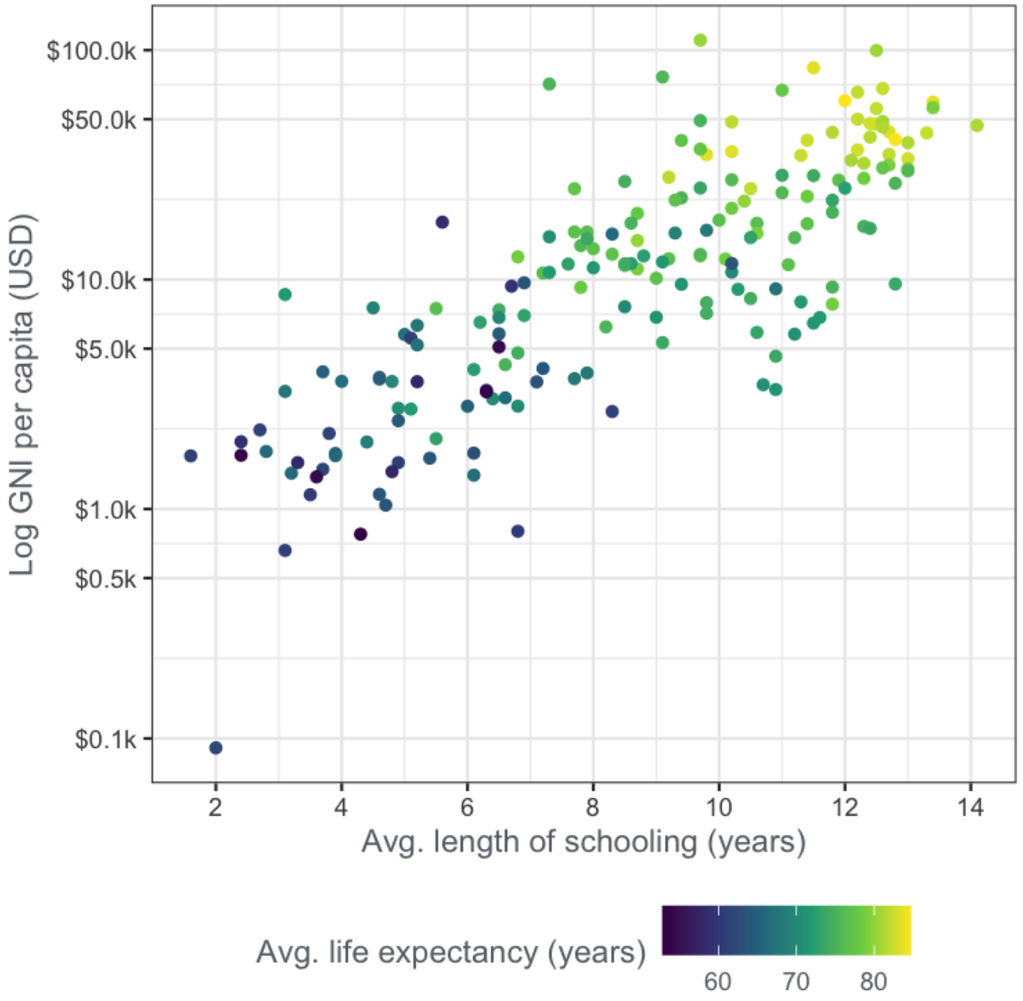
# When should you use a correlation heatmap?

- You have lots of continuous variables.
- You want to a simple overview of how each pair of variables is related.




[1] Rossi, Allenby, and McCulloch (2005). Bayesian Statistics & Marketing

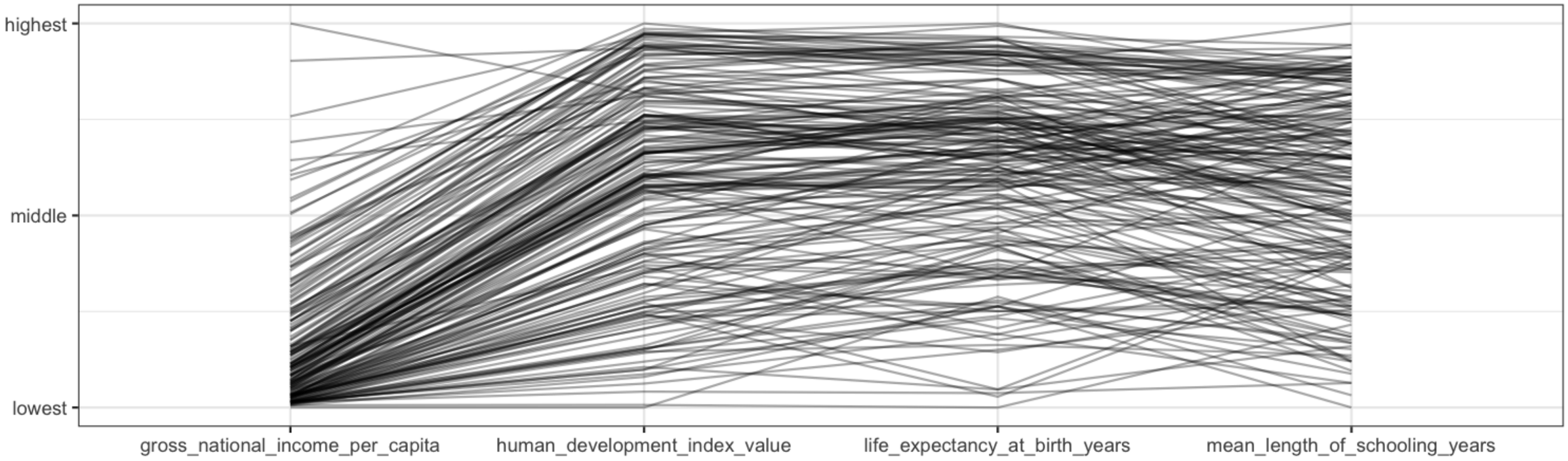
# The United Nations dataset again

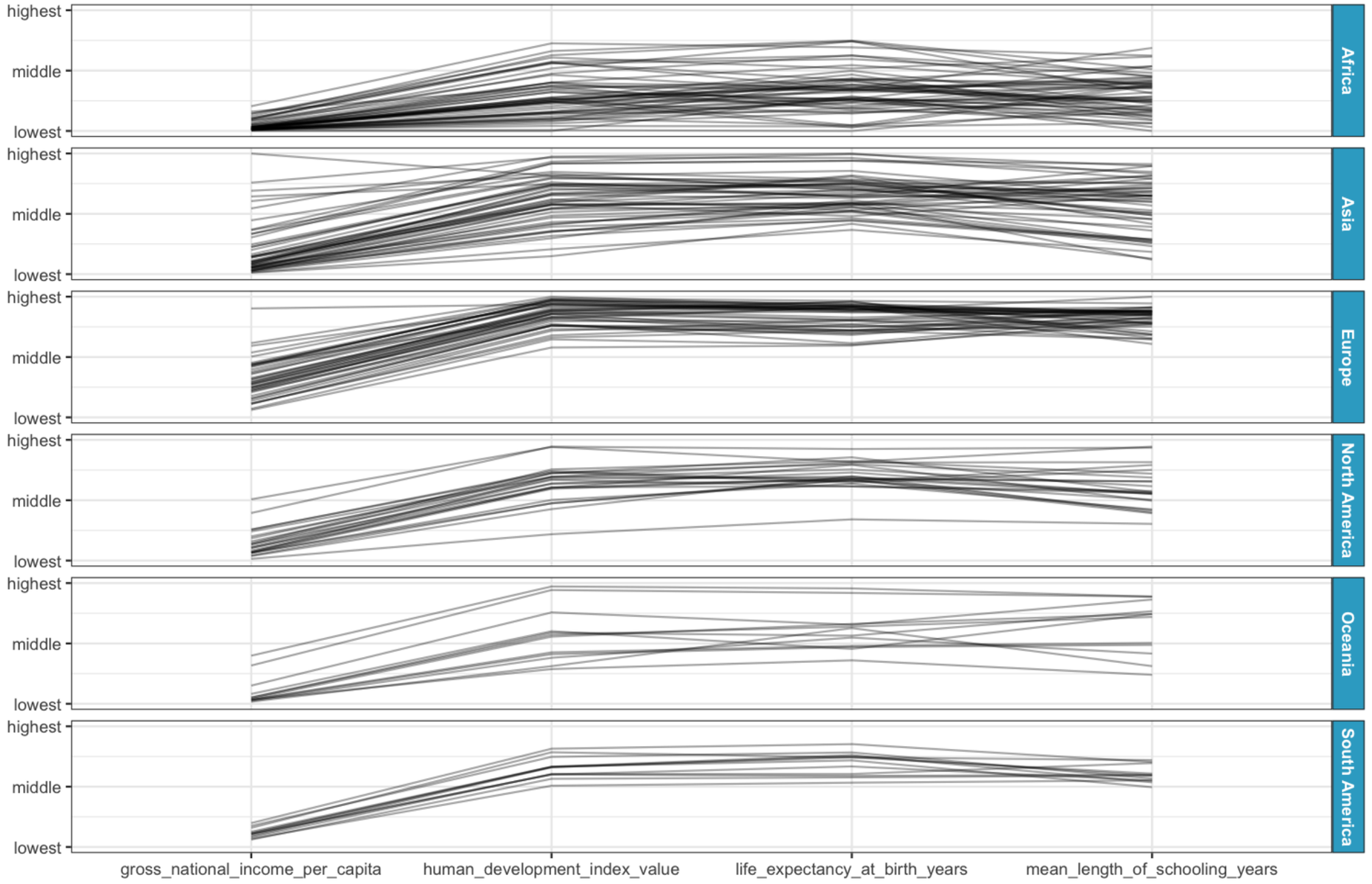


# When should you use a parallel coordinates plot?

- You have lots of continuous variables.
- You want to find patterns across these variables, or
- You want to visualize clusters of observations.

# A parallel coordinates plot




highest
middle
lowest
Africa
Asia
Europe
North America
Oceania
South America
gross_national_income_per_capita
human_development_index_value
life_expectancy_at_birth_years
mean_length_of_schooling_years

# Let's practice!

UNDERSTANDING DATA VISUALIZATION